

Experiment 8: Naive Bayes Classifier

Title: Implementation of Naive Bayes Classifier using Python

Objective: To understand and implement the Naive Bayes algorithm for classification using a sample dataset.

Theory: Naive Bayes is a probabilistic classifier based on Bayes' Theorem with a strong independence assumption between features. It is especially suited for large datasets and works well in text classification tasks like spam detection.

Algorithm

1. Convert the dataset into frequency tables.
2. Create a likelihood table using probabilities.
3. Use Bayes' theorem to calculate posterior probability for each class.
4. Assign the class with the highest posterior probability to the input.

Code (Illustration with Naïve Bayes)

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, classification_report

# Load dataset
iris = load_iris()
X_train, X_test, y_train, y_test = train_test_split(iris.data, iris.target, test_size=0.3,
random_state=42)

# Train Naive Bayes model
model = GaussianNB()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Evaluate
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```

Input

- Dataset: Iris dataset
- Classifier: Gaussian Naïve Bayes

Output

- Accuracy score
- Classification report

Analysis

Naive Bayes is simple yet effective, particularly for classification tasks with text or categorical data. Its assumption of independence among features might not always be realistic, but the algorithm performs well in practice.

Conclusion

Naive Bayes is a fast and efficient classification algorithm based on probabilistic learning, useful in applications like spam filtering, sentiment analysis, and document categorization.

Viva Questions

1. What is Bayes' Theorem?
2. What is the assumption behind Naive Bayes?
3. What are the types of Naive Bayes classifiers?
4. How does Gaussian Naive Bayes differ from Multinomial Naive Bayes?
5. When should you use Naive Bayes?
6. What are the limitations of Naive Bayes?
7. How do you handle zero probabilities in Naive Bayes?
8. What is the mathematical formula for posterior probability?
9. Calculate the posterior probability manually given prior, likelihood, and evidence.
10. How does Naive Bayes perform when features are correlated?
11. Can Naive Bayes be used for regression problems?
12. Explain Laplace smoothing and its importance in Naive Bayes.
13. Derive the formula for Bayes' Theorem.
14. How would the performance of Naive Bayes differ with continuous vs categorical features?
15. What datasets are best suited for Naive Bayes?
16. **Mathematical:** Given the prior probabilities $P(A) = 0.4$ and $P(B) = 0.6$, and likelihoods $P(x|A) = 0.5$, $P(x|B) = 0.2$, calculate $P(A|x)$ and $P(B|x)$.
17. **Mathematical:** Given mean ($\mu = 5$) and standard deviation ($\sigma = 2$) for a Gaussian distribution, calculate $P(x=6)$ using the Gaussian Naive Bayes formula.
18. **Mathematical:** For two classes $C1$ and $C2$, and a binary feature vector $x = [1, 0]$, compute $P(C1|x)$ and $P(C2|x)$ using naive Bayes assumptions and given probabilities.
19. **Mathematical:** Explain the impact of log transformation in probability multiplication in Naive Bayes.
20. **Mathematical:** If you are given class conditional probabilities and prior probabilities, how do you normalize them to get final posterior values?